Introduction to Graph Theory

5.1 ⇒ Directed Graphs (Digraph)

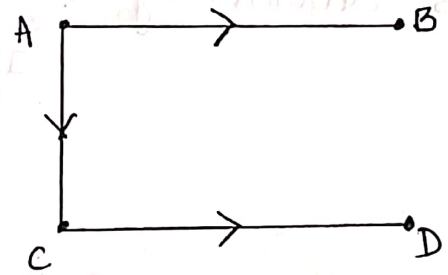
A directed graph (or a digraph) is a pair (V, E), where V is a nonempty set and E is a set of ordered pairs of elements taken from the set V.

It is represented as :

$G = (V, E)$ (or) $D = (V, E)$

* V = Set of Vertices (points or nodes)
* E = Set of Ordered pairs of vertices called 'directed edges'

Ex :



* Vertex set, $V = \{A, B, C, D\}$
* Edge set, $E = \{AB, CD, CA\}$
$= \{(A,B), (C,D), (C,A)\}$

⇒ Initial Vertex

The initial vertex is the starting point of a directed edge in a directed graph.

⇒ Terminal Vertex (Final vertex)

The Terminal vertex is the ending point of a directed edge in a directed graph.

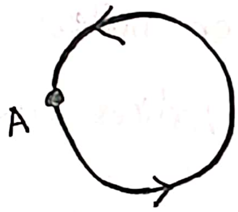
Ex :

* A → Initial vertex
* B → Terminal (final) vertex

→ **Directed Loop**

A directed loop is a directed edge that starts and ends at the same vertex.

Ex :



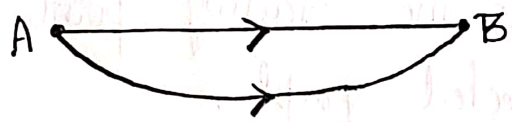
A directed loop 'A' is denoted by, $A = (A, A)$

$\downarrow$ Initial vertex, $\downarrow$ Terminal vertex

→ **Parallel directed edges.**

Two directed edges having the same initial vertex and the same terminal vertex are called 'Parallel directed edges'.

Ex :



⇒ Multiple directed edges

Two or more directed edges having the same initial vertex and the same terminal vertex are called 'multiple directed edges'.

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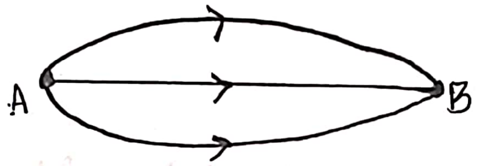


⇒ Isolated vertex

A vertex having no incoming and no outgoing edges is called an isolated vertex.

Ex : A.

⇒ Non-isolated vertex

A non-isolated vertex is a vertex that has at least one edge connected to it.

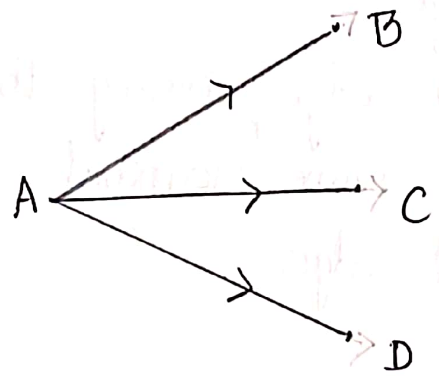
Ex :

A and B are non-isolated vertices.

⇒ Source

A source is a vertex in a directed graph that has only outgoing edges and no incoming edges.

Ex :

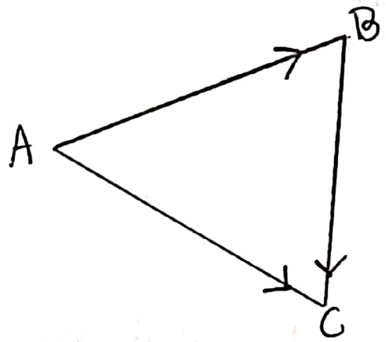


Here 'A' is a Source

⇒ Sink :

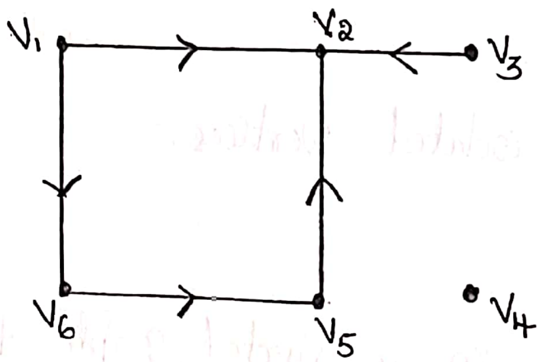
A Sink is a Vertex in a directed graph that has only incoming edges and no outgoing edges.

Ex :



Here 'C' is a Sink

⇒ Isolated, Non-isolated, Source and sink in one diagram.



V_4 → Isolated vertex

V_1 & V_3 → Sources

V_2 → Sink

V_5 & V_6 → do not belong to any of these Categories.

In-degree and out-degree

⇒ **In-degree :**

In-degree of a vertex in a directed graph is the number of directed edges that enter at that vertex.

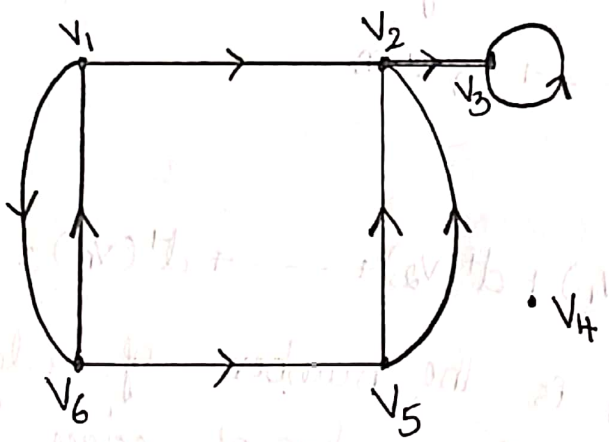
It is denoted by $deg^{-}(v)$.

⇒ **out-degree :**

Outdegree of a vertex in a directed graph is the number of directed edges that leave from that vertex.

It is denoted by $deg^{+}(v)$

Ex :



In-degree(v)	out-degree(v)
$d^{-}(v_1) = 1$	$d^{+}(v_1) = 2$
$d^{-}(v_2) = 3$	$d^{+}(v_2) = 1$
$d^{-}(v_3) = 2$	$d^{+}(v_3) = 1$
$d^{-}(v_4) = 0$	$d^{+}(v_4) = 0$
$d^{-}(v_5) = 1$	$d^{+}(v_5) = 2$
$d^{-}(v_6) = 1$	$d^{+}(v_6) = 2$

∴ If v is a sink, then $d^{-}(v)$ & $d^{+}(v) = 0$

Property: In every digraph D, the sum of the out-degrees of all vertices is equal to the sum of the in-degrees of all vertices, each sum being equal to the number of edges in D.

Proof: Suppose D has n vertices $v_1, v_2, ---, v_n$ and m edges.

Let r_1 be the number of edges going out of v_1, r_2 be the number of edges going out of v_2, and so on.

Then, $d^+(v_1) = r_1$, $d^+(v_2) = r_2$, $---$, $d^+(v_n) = r_n$.

Since every edge terminates at some vertex and since there are m edges, we should have

$$r_1 + r_2 + --- + r_n = m$$

Accordingly,

$$d^+(v_1) + d^+(v_2) + --- + d^+(v_n) = r_1 + r_2 + --- + r_n = m.$$

Similarly, if S_1 is the number of edges coming into v_1, S_2 is the number of edges coming into v_2, and so on, we get

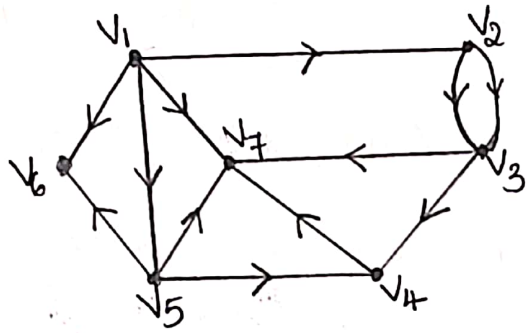
$$d^-(v_1) + d^-(v_2) + --- + d^-(v_n) = S_1 + S_2 + --- + S_n = m.$$

Thus,

$$\sum_{i=1}^{n} d^+(v_i) = \sum_{i=1}^{n} d^-(v_i) = m$$

Hence prooved

Example 1 : Find the in-degrees and out-degrees of the vertices of the given digraph. and also check that sum of out-degrees is equal to sum of in-degrees = No. of edges.



Soln

Given : No. of vertices = 7
No. of edges = 12

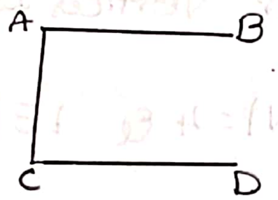
In-degree	out-degree
$d^-(v_1) = 0$	$d^+(v_1) = 4$
$d^-(v_2) = 1$	$d^+(v_2) = 2$
$d^-(v_3) = 2$	$d^+(v_3) = 2$
$d^-(v_4) = 2$	$d^+(v_4) = 1$
$d^-(v_5) = 1$	$d^+(v_5) = 3$
$d^-(v_6) = 2$	$d^+(v_6) = 0$
$d^-(v_7) = 4$	$d^+(v_7) = 0$
$\sum_{i=1}^{7} d^-(v_i) = 12$	$\sum_{i=1}^{7} d^+(v_i) = 12$

$\therefore$ Sum of out-degrees = Sum of in-degrees = 12 = No. of edges.

5.2- Graph.

A graph is a pair (V, E) where V is a nonempty set and E is a set of ordered pair of elements taken from the set V.

The graph (V, E) is denoted by G = (V, E) or G(V, E)



G = G(4,3)

Note ① In (V, E), V is Vertex set & E is edge set

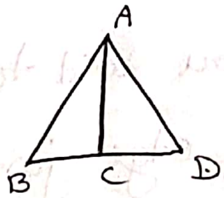
② A graph containing no edges is called null graph
[In null graph every Vertex is an isolated Vertex]

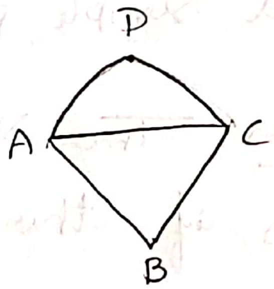
③ Null graph with only one Vertex is called trivial graph.

④ A graph with finite no. of Vertex & edges. is called finite graph otherwise infinite graph.

⑤ In graph edges may be straight line or curve

Ex: G(4, 5)





Order and Size

The no. of Vertices in a graph is called the order of graph (|V|) & the no. of edges in it is called its size (|E|)

A graph of order 'm' & size 'n' is called (m, n) graph

A null graph with 'n' Vertices is (n, 0) graph

Ex. $G = G(4, 10)$ means $|V| = 4$ & $|E| = 10$

Loop: An edge having the same initial & final Vertex.

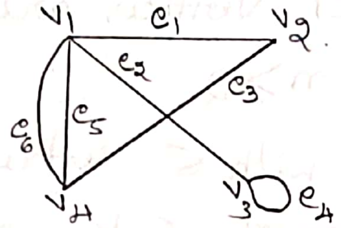
Parallel edges: Two edges having the same end Vertices.

Multiple edges. Two or more edges with the same initial and end vertices.

Adjacency: Two Vertices are adjacent if connect-ed Vertices; a simple path has no repeated Verti-ces.

Adjacent Vertices. Two Vertices are said to be adjacent Vertices if there is an edge joining b/w them.

Let v_i & v_j denotes the two vertices of a graph and e_k denotes an edge joining v_i & v_j then v_i & v_j are called end vertices (end points) of e_k. This is symbolically written as $e_k = \{v_i, v_j\} = v_i v_j$



In this figure $e_1 = \{v_1, v_2\} = v_1 v_2$

$e_2 = \{v_1, v_3\} = v_1 v_3$

$e_3 = \{v_2, v_4\} = v_2 v_4$

$e_4 = \{v_3, v_3\} = v_3 v_3$

$e_5 = \{v_1, v_4\} = v_1 v_4$

$e_6 = \{v_1, v_4\} = v_1 v_4$

Simple graph. A graph doesnot contain loops and multiple edges is called simple graph.

Loop free graph. A graph does not contain a loop is called loop graph.

Multigraph. A graph which contains multiple edges but no loop is called multigraph.

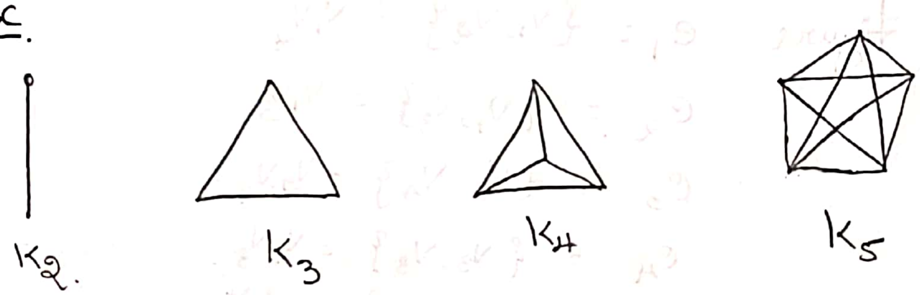
General graph. A graph which contains multiple edges (or) loops (or) path is called general graph.

Complete graph. A simple graph of order $n \geq 2$ in which there is an edge b/w every pair of vertices is called a Complete graph (or) full graph.

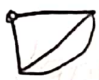
Complete graph is simple graph in which every pair of distinct vertices are adjacent. It is denoted by K_n, $n \geq 2$.

Complete graph with 5 vertices is called Kuratowski's first graph (K_5-graph)

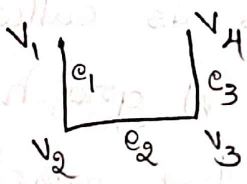
Ex.



Note ① Name is not assigned to vertices of a graph is called unlabeled graph.



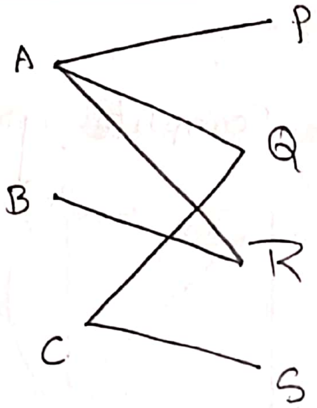
② Name is assigned to vertices of a graph is called labeled graph.



Bipartite graph. A graph $G = (V, E)$ is called bipartite if $V = V_1 \cup V_2$ with $V_1 \cap V_2 = \phi$ & every edge of G is of the form $\{a, b\}$ with $a \in V_1$ & $b \in V_2$.

Bipartite graph can be denoted by $G = (V_1, V_2 : E)$

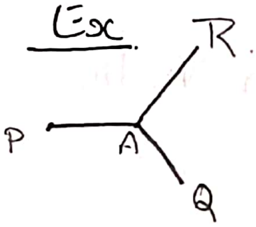
The sets V_1 & V_2 are called bipartites (partition) of the vertex set V



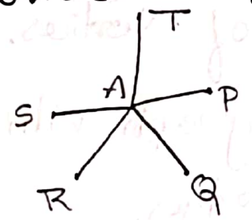
This is bipartite graph.
here $V_1 = \{A, B, C\}$ &
$V_2 = \{P, Q, R, S\}$ are bipartites

Complete bipartite graph. A bipartite graph $G = (V_1, V_2 : E)$ is called called complete bipartite graph, if there is an edge b/w every vertex from V_1 to every vertex in V_2.

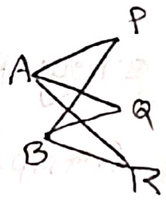
Ex.



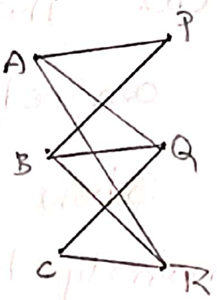
$K_{1,3}$



$K_{1,5}$



$K_{2,5}$



$K_{3,3}$

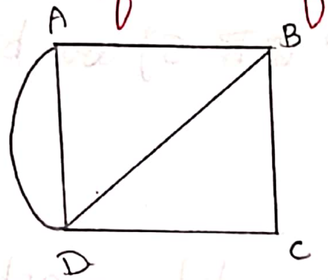
Note. ① $K_{3,3}$ graph is called Kuratowski's second graph.

② A bipartite graph G is not a complete graph even if G is a complete bipartite graph. Because in such a graph there exists no edge b/w two vertices if they belong to the same bipartite.

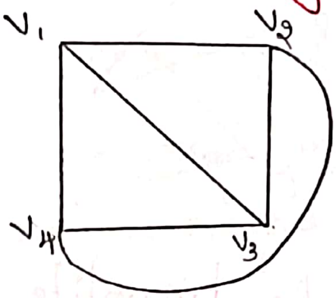
Problems

① Which of the following is a Complete graph?

(a)



(b)



Soln

(a) This graph is not a Complete graph.
$\because$ there is no edge b/w A & C.

(b) This graph is Complete graph. $\because$ there is an edge b/w every pair of Vertices.

② Show that a Complete graph with n Vertices namely K_n has $\frac{1}{2}n(n-1)$ edges.

Proof: In a Complete graph there exist exactly one edge b/w every pair of Vertices.
i.e the no of edges in complete graph is equal to no of pairs of Vertices.
If the no of Vertices is n then the no of pairs of Vertices is $^nC_2 = \frac{n!}{(n-2)!\,2!} = \frac{1}{2}n(n-1)$

Thus the no of edges in a complete graph with n Vertices is $\frac{1}{2}n(n-1)$.

③ If $G = G(V,E)$ is a Simple graph Prove that
$$2|E| \leq |V|^2 - |V|$$

Proof Each edge of a graph is determined by a pair of Vertices.
In a Simple graph there is no multiple edges.

In a simple graph, the nor of edges cannot exceed the nor of pair of vertices.

The nor of pair of vertices that can be choosen from 'n' vertices is $^nC_2 = \frac{1}{2}n(n-1)$.

Thus for simple graph with $n \geq 2$ vertices the nor of edges cannot exceed $\frac{1}{2}n(n-1)$ edges

In a simple graph $G = G(V, E)$ has 'n' vertices & 'm' edges then $m \leq \frac{1}{2}n(n-1)$

$$2m \leq (n^2 - n)$$

$$\text{i.e } 2|E| \leq |V|^2 - |V|.$$

(4) Show that a simple graph of $n=4$ & size $m=7$ and a complete graph of order $n=4$ & size $m=5$ do not exist

Sol^n For $n=4$ we have $\frac{1}{2}n(n-1) = \frac{1}{2}(4)(3) = 6$

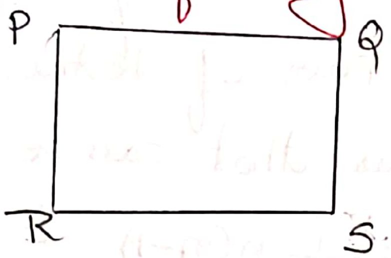
Since $m=7$ exceeds this nor, a simple graph of order $n=4$ & size $m=7$ doesnot exist.

Similarly since $m=5$ is not equal to $6 = \frac{1}{2}n(n-1)$

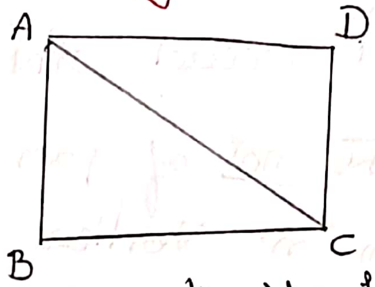
$\therefore$ A complete graph of order 4 & size $m=5$ doesnot exists.

(5) Which of the following is a bipartite graph?

(a)



(b)



Solⁿ: (a) This is bipartite graph with $V_1 = \{P, S\}$ &
$V_2 = \{Q, R\}$ are the bipartites.

(b). This graph is not a bipartite graph.

(6) (a) How many Vertices & how many edges are there in the complete bipartite graph $K_{4,7}$ & $K_{7,11}$?

(b). If the graph $K_{n,12}$ has 72 edges, what is n?

Solⁿ W.K.T the complete bipartite graph $K_{n,s}$ has $(n+s)$ Vertices & ns edges.

(a) ∴ The graph $K_{4,7}$ has $4+7 = 11$ Vertices & $4 \times 7 = 28$ edges and the graph $K_{7,11}$ has 18 Vertices & 77 edges.

(b) If the graph $K_{n,12}$ has 72 edges.

$$\therefore 12n = 72.$$

$$n = 6$$

Subgraphs. G_1 is a subgraph of G if.

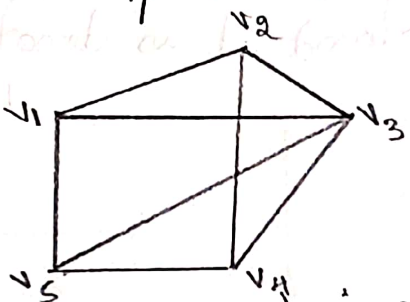
i) All the Vertices and all the edges of G_1 are in G

ii) Each edge of G_1 has the same end vertices in G.

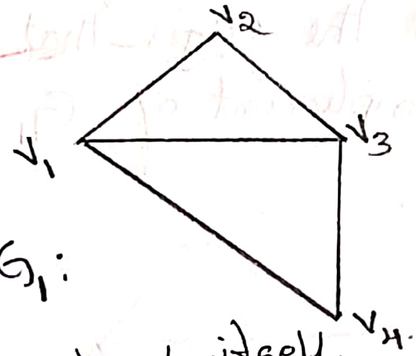
as in G_1

Ex.

G:



G_1:



Note: ① Every graph is a subgraph of itself.

② Every simple graph of n vertices is a subgraph of the complete graph K_n.

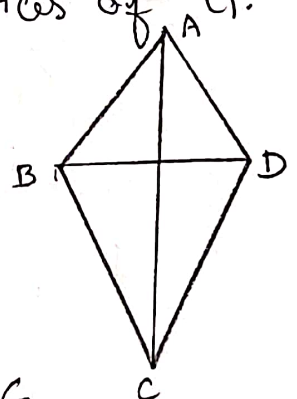
③ If G_1 is a subgraph of a graph G_2 & G_2 is a subgraph of a graph G, then G_1 is a subgraph of G

i.e, $G_1 \subseteq G_2$ & $G_2 \subseteq G$ then $G_1 \subseteq G$.

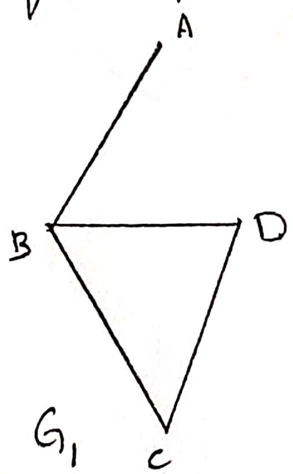
④ A single vertex in a graph G is a subgraph of G.

⑤ A single edge in a graph G together with its end vertices is a subgraph of G.

Spanning Subgraph. A Subgraph G_1 of a graph. G is a spanning subgraph of G if G_1 contains all vertices of G.



G.

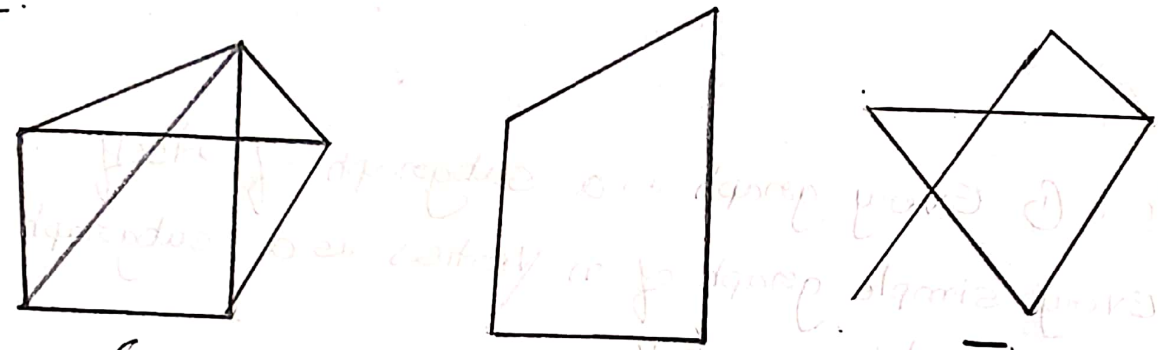


G_1

Complement of a Subgraph.

Let G be a graph & G_1 is subgraph of G, the subgraph of G obtained by deleting from G all the edges that belong to G_1 is called the complement of G_1 in G and it is denoted by $\overline{G_1}$.

Ex.



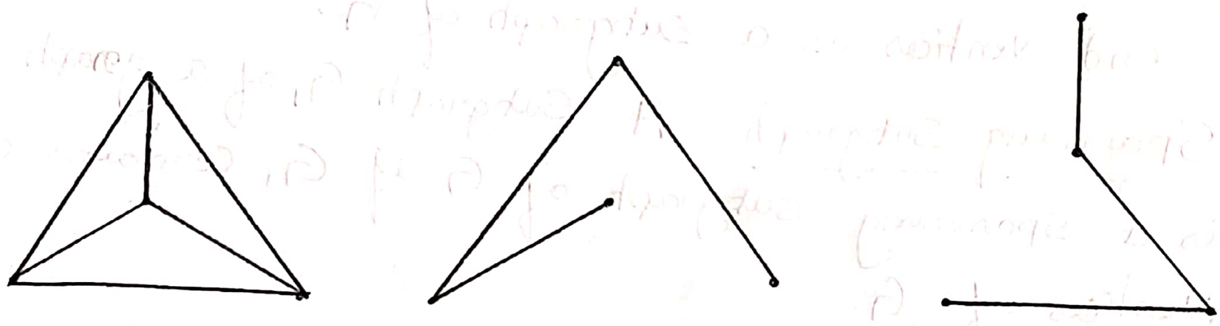
G $\qquad$ G_1 $\qquad$ $\overline{G_1}$

Complement of a simple graph.

If G is a simple graph of order n, then the complement of G in K_n is called the complement of G & it is denoted by $\overline{G}$.

i.e, $\overline{G} = K_n - G$

Ex



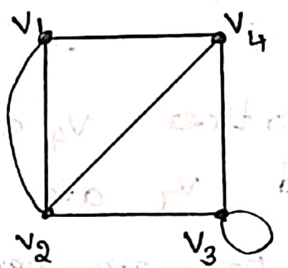
K_4 $\qquad$ G $\qquad$ $\overline{G}$

5.3 Vertex Degree and Handshaking Property

Let $G=(V,E)$ be a graph and v be a vertex of G. Then, the number of edges of G that are incident on v with the loops counted twice is called the "degree of the vertex v" and is denoted by $\deg(v)$, (or) $d(v)$.

The degrees of all vertices in a graph arranged in non-decreasing order are called the "degree sequence" of the graph. Also the minimum of the degrees of vertices of a graph is called the "degree of graph."

Ex:



$d(v_1)=3$, $d(v_2)=4$, $d(v_3)=4$, $d(v_4)=3$.

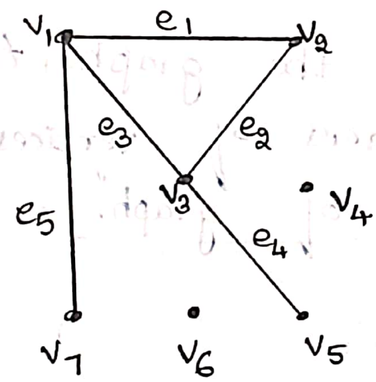
Therefore, the degree sequence of this graph is 3, 3, 4, 4 and the degree of the graph is 3. Note that the loop at v_3 is counted twice for determining the degree of v_3.

Isolated vertex, pendant Vertex :-

A vertex in a graph which is not an end vertex of any edge of the graph is called an "isolated vertex". obviously, a vertex is an isolated vertex if and only if its degree is zero.

A vertex of degree 1 is called a pendent vertex. An edge incident on a "pendent vertex" is called a "pendent edge".

Ex:



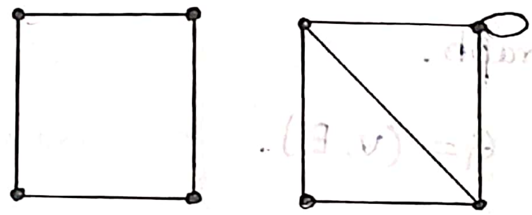
In the graph the vertices v_4 and v_6 are isolated vertices, v_5 and v_7 are pendant vertices and the edges e_4 and e_5 are pendant edges.

As mentioned before, a null graph contains no edges. It therefore follows that in a nullgraph every vertex is an isolated vertex.

Regular graph :-

A graph in which all the vertices are of the same degree 'k' is called a "regular graph of degree k, (or) a k-regular graph."

In particular, a 3-regular graph is called a "cubic graph".

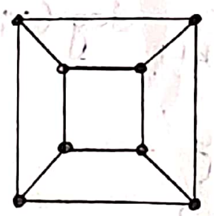


2-regular graph   4-regular graph

The 3-regular graph is a cubic graph, which contains 10 vertices and 15 edges, is called the petersen graph.

3-regular graph

For any positive integer k, a loop-free k-regular graph with 2^k vertices is called the k-dimensional hypercube (or k-cube) and is denoted by Q_k.



Graph G

The graph G is a cubic graph with $2^3 = 8$ vertices. This graph is known as "three-dimensional hypercube" and is denoted by Q_3.

Handshaking Property

The sum of the degrees of all the vertices in a graph is an even number; and this number is equal to twice the number of edges in the graph.

ie For a graph $G = (V, E)$.

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Theorem :- In every graph, the number of vertices of odd degrees is even.

Proof:- Let a graph contain n vertices and k of these vertices are of odd degrees, then the remaining $n-k$ vertices are of even degree.

Let the vertices with odd degree be $v_1, v_2 \ldots v_k$ and the vertices with even degree be $v_{k+1}, v_{k+2} \ldots v_n$.

Sum of the degree of the vertices is,

$$\sum_{i=1}^{n} \deg(v_i) = \sum_{i=1}^{k} \deg(v_i) + \sum_{i=k+1}^{n} \deg(v_i) \longrightarrow (1)$$

By handshaking property

$$\sum_{i=1}^{n} \deg(v_i) = 2|E| \longrightarrow (2)$$

Hence from (1) and (2)

$$\Rightarrow \sum_{i=1}^{k} deg(v_i) + \sum_{i=k+1}^{n} deg(v_i) = \text{Even}$$

The Second term in the LHS is the sum of the degree of vertices with even degrees, this sum is also even. Therefore the first term in LHS must be even.

$$\Rightarrow \sum_{i=1}^{k} deg(v_i) = \text{Even}$$

ie $d(v_1) + d(v_2) + \dots + d(v_k) = \text{even}$.

$\therefore$ The number of vertices of odd degree is even.

Example 1:

Can there be a graph with 12 vertices such that two of the vertices have degree 3 each and the remaining 10 vertices have degree 4 each?

Solution:

Here, the sum of the degrees of vertices is $(3\times2)+(4\times10)=46$.

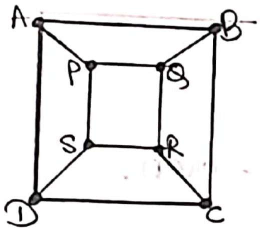
$\therefore$ If $m=23$, we have $2m=46$ and the handshaking property holds.

Hence there can be a graph of the desired type whose size is 23.

Problems

1. Show that the hypercube Q_3 is a bipartite graph which is not a complete bipartite graph.

Solution: The hypercube (Q_3) is a graph with the vertex set $V = \{A, B, C, D, P, Q, R, S\}$



Let $V_1 = \{A, C, Q, S\}$ and $V_2 = \{B, D, P, R\}$ be the two vertex set of Q_3.

From the graph, we note that

(i) $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \phi$

(ii) Every edge of the graph has one end vertex in the set V_1 and the other end vertex in the set V_2. Also no edge of the graph as both of its end vertices in V_1 or V_2.

Hence hypercube Q_3 is a bipartite graph.

Hypercube Q_3 is not a complete bipartite graph as there is no edge between every vertex in V_1 to every vertex in V_2.

[e.g: there is no edge from A to R].

2. Prove that the k-dimensional hypercube Q_k has $k2^{k-1}$ edges. Determine the number of edges in Q_8. Find the dimension of the hypercube with 524288 edges?

Solution:- In Q_n, the number of vertices is 2^n and each vertex is of degree n

∴ The sum of the degree of vertex of Q_n is

$$\sum_{i=1}^{n} \deg(v_i) = n \times 2^n$$

By handshaking property, $n \times 2^n = 2|E|$

ie $|E| = n \cdot 2^{n-1}$

The number of edges for Q_8 is $8 \times 2^7 = 1024$.

In k-dimensional hypercube Q_k, the number of vertices is 2^k and the number of edges is $k2^{k-1}$.

If Q_k has 524288 edges,

ie $k2^{k-1} = 524288$

$\Rightarrow k2^{k-1} = 524288 = 2^{19}$

$\Rightarrow k2^{k-1} = 2^4 \times 2^{15}$

$\Rightarrow k2^{k-1} = 16 \times 2^{15}$ ——— (1)

Equation (1) holds if $k = 16$

Thus, the dimension of the hypercube with 524288 edges is $k = 16$.

3. Determine the order $|V|$ of the graph $G = (V, E)$ in the following cases:

(i) G is a cubic graph with 9 edges.

(ii) G is regular with 15 edges.

(iii) G has 10 edges with 2 vertices of degree 4 and all other vertices of degree 3.

Solution :-

(i) Since G is a cubic graph with n vertices, all the n vertices of G has degree 3

and $E = m = 9$

$$\sum_{i=1}^{n} \deg(v_i) = 3 \times n$$

By Handshaking property $3n = 2 \times 9$

$3n = 18$

$\boxed{n = 6}$

(ii) Given:

G is regular with 15 edges, ie $m = 15$.

$\Rightarrow$ all vertices of G have the same degree (k) and n be the number of vertices.

$$\sum_{i=1}^{n} \deg(v_i) = nk.$$

By Handshaking property $\sum_{i=1}^{n} \deg(v_i) = 2m$

$\Rightarrow nk = 2 \times 15$

$\Rightarrow k = \frac{30}{n}$.

Since k is a positive integer

$\Rightarrow$ n must be a positive divisor of 30.

$\therefore$ order (n) of G are 1, 2, 3, 5, 6, 10, 15, 30

(iii) Given:

G has 10 edges with 2 vertices of degree 4 and all other vertices of degree 3.

$\Rightarrow$ 2 vertices of G are of degree 4 and all other of degree 3,

$m=10$ and n be the number of vertices in G.

The sum of degree is $\sum_{i=1}^{n} deg(v_i) = 2\times4+(n-2)\times3$

By Handshaking property $\sum_{i=1}^{n} deg(v_i) = 2m$

$$\Rightarrow 8+3n-6 = 2\times10$$

$$\Rightarrow 3n+2 = 20$$

$$\Rightarrow \cancel{3}n = \cancel{18}\,6$$

$$\Rightarrow \boxed{n=6}$$

4. Show that there is no graph with 28 edges and 12 vertices in the following cases:

(i) The degree of a vertex is either 3 (or) 4.

(ii) The degree of a vertex is either 3 (or) 6.

Solution:-

Suppose there is a ~~path~~ graph with 28 edges and 12 vertices, of which k vertices are of degree 3.

Then: (i) If all of the remaining $(12-k)$ vertices have degree 4, $m=28$

By Handshaking property $\sum_{i=1}^{n} deg(v_i) = 2m$

$$\Rightarrow 3k+4(12-k) = 2\times 28$$
$$\Rightarrow 3k+48-4k=56$$
$$\Rightarrow -k = 56-48$$
$$\Rightarrow k=-8$$

which is not possible (Because k has to be non-negative).

(ii) If all of the remaining $(12-k)$ vertices have degree 6, $m=28$

By Handshaking property $\sum_{i=1}^{n} \deg(v_i) = 2m$

$$\Rightarrow 3k+6(12-k) = 2\times 28$$
$$\Rightarrow 3k+72-6k=56$$
$$\Rightarrow -3k=56-72$$
$$\Rightarrow -3k=-16$$
$$\Rightarrow k=16/3$$

which is not possible (Because k must be an integer).

5. (a) If a graph with n vertices and m edges is k-regular, show that $m=kn/2$.

(b) Does there exist a cubic graph with 11 vertices?

(c) Does there exist a 4-regular graph with (i) 15 edges? (ii) 10 edges?

Solution:-

(a) G has n vertices and m edges and is k-regular (degree of every vertex is k)

$$\therefore \sum_{i=1}^{n} \deg(v_i) = nk$$

By Handshaking property, $nk = 2m$

$$\Rightarrow m = \frac{nk}{2}$$

(b) G is cubic graph with $n=11$ vertices and $k=3$

By Handshaking property $\sum_{i=1}^{n} \deg(v_i) = 2m$

$$3 \times 11 = 2m$$

$$\therefore m = \frac{3\times 11}{2} = \frac{33}{2}$$

This is not possible, as edges m must be an integer.

Hence, the graph doesn't exist.

(c) (i) G is a 4-regular graph with $m=15$ vertices and $k=4$

By Handshaking property $\sum_{i=1}^{n} \deg(v_i) = 2m$

$$4 \times n = 2\times 15 = 30$$

$$\therefore n = \frac{\cancel{30}}{\cancel{4}} = \frac{15}{2}$$

This is not possible, as vertices n must be an integer.

Hence, the graph doesn't exist.

(ii) G is a 4-regular graph with $m=10$ vertices and $k=4$

By Handshaking property $\sum_{i=1}^{n} \deg(v_i) = 2m$

$$4\times n = 2 \times 10 = 20$$

$$\cancel{4} \times n = \cancel{20}^{5}$$

$$\therefore n = 5$$

Hence, the graph exist.

6. For a graph with n vertices and m edges, if δ is the minimum and Δ is the maximum of the degrees of vertices, show that $\delta \leq \frac{2m}{n} \leq \Delta$.

Proof:- Let $d_1, d_2, d_3 \dots d_n$ be the degrees of the vertices. Then, by Handshaking property, we have

$$d_1 + d_2 + d_3 + \dots + d_n = 2m. \quad \text{———— (i)}$$

Since $\delta = \min(d_1, d_2, d_3 \dots d_n)$, we have

$$d_1 \geq \delta, \; d_2 \geq \delta, \; \dots \; d_n \geq \delta.$$

Adding these n inequalities, we get

$$d_1 + d_2 + d_3 + \dots + d_n \geq n\delta. \quad \text{———— (ii)}$$

Similarly, Since $\Delta = \max(d_1, d_2, d_3 \dots d_n)$, we get

$$d_1 + d_2 + d_3 + \dots + d_n \leq n\Delta. \quad \text{———— (iii)}$$

From (i), (ii) and (iii),

we get $2m \geq n\delta$ and $2m \leq n\Delta$,

so that $n\delta \leq 2m \leq n\Delta$,

$$\delta \leq \frac{2m}{n} \leq \Delta.$$

5.4 – ISOMORPHISM

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two groups then a function $f: V_1 \to V_2$ is called a graph isomorphism if (i) f is one-to-one and onto (one to one correspondence)

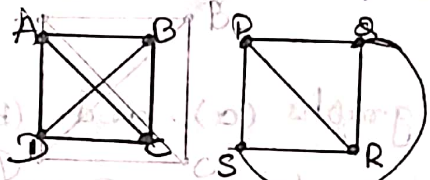
(ii) For all $a, b \in V_1$, $(a, b) \in E_1$ iff $\{f(a), f(b)\} \in E_2$.

when such a function exists G_1 and G_2 are called isomorphic groups.

Example 1:- The two graphs G_1 and G_2 are isomorphic. Since there is a one to one correspondence between the vertices / edges of the two graphs preserve the vertices.

The corresponding vertices of G_1 and G_2 are

$A \leftrightarrow P$, $B \leftrightarrow Q$, $R \leftrightarrow C$, $D \leftrightarrow S$

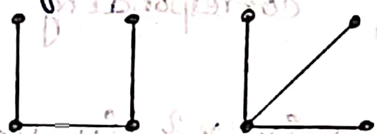


And the corresponding edges are

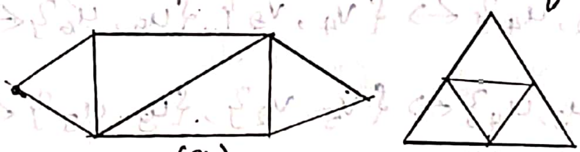
$\langle A,B\rangle \leftrightarrow \langle P,Q\rangle$, $\langle B,C\rangle \leftrightarrow \langle Q,R\rangle$, $\langle A,D\rangle \leftrightarrow \langle P,S\rangle$,

$\langle D,C\rangle \leftrightarrow \langle S,R\rangle$, $\langle A,C\rangle \leftrightarrow \langle P,R\rangle$, $\langle B,D\rangle \leftrightarrow \langle Q,S\rangle$.

Note:- Two graphs of the same order and the same size need not be isomorphic.



Example: 2. Show that the following graphs are not isomorphic.



(a) (b)

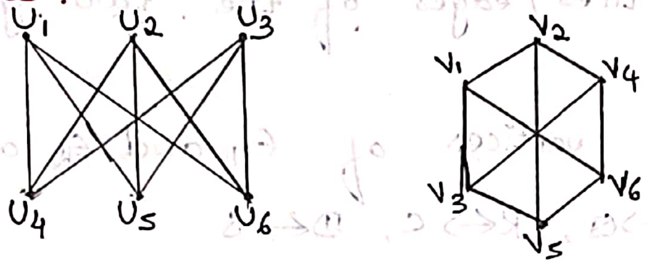
Each of the two graphs has 6 vertices and 9 edges. But, the first graph has 2 vertices of degree 4 whereas the second graph has 3 vertices of degree 4.

$\therefore$ There cannot be any one-to-one correspondence between the vertices and between the edges of the two graphs which preserves the adjacency of vertices.

$\therefore$ The two graphs are not isomorphic.

Problems

1. Show that the following two graphs are isomorphic:



Solution:- Both the graphs (a) and (b) have six vertices each of degree 3 and 9 edges.

The corresponding vertices are

$u_1 \leftrightarrow v_1$, $u_2 \leftrightarrow v_4$, $u_3 \leftrightarrow v_5$,

$u_4 \leftrightarrow v_2$, $u_5 \leftrightarrow v_3$, $u_6 \leftrightarrow v_6$.

These yields the following corresponding between the edges:

$\{u_1, u_4\} \leftrightarrow \{v_1, v_2\}$, $\{u_1, u_5\} \leftrightarrow \{v_1, v_3\}$, $\{u_1, u_6\} \leftrightarrow \{v_1, v_6\}$,

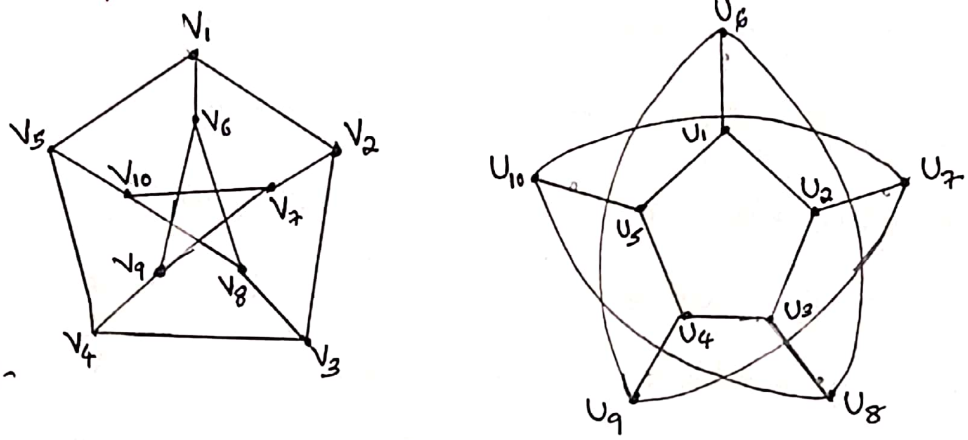
$\{u_2, u_5\} \leftrightarrow \{v_4, v_3\}$, $\{u_2, u_4\} \leftrightarrow \{v_4, v_2\}$, $\{u_2, u_6\} \leftrightarrow \{v_4, v_6\}$

$\{u_3, u_6\} \leftrightarrow \{v_5, v_6\}$, $\{u_3, u_4\} \leftrightarrow \{v_5, v_2\}$, $\{u_3, u_5\} \leftrightarrow \{v_5, v_3\}$

The above correspondences between the edges and the vertices are one-to-one correspondences and these preserve the adjacency of vertices.

Hence the two graphs are isomorphic.

2. Show that the following two graphs are isomorphic.



Solution:-

Both the graphs (a) and (b) have ten vertices each of degree 3 and 15 edges.

The corresponding vertices are $V_i \leftrightarrow u_i$ where $i = 1$ to 10.

These yield the following corresponding edges are $\{v_i, v_j\} \leftrightarrow \{u_i, u_j\}$ $i \neq j$ where $i = 1$ to 10 & $j = 1$ to 10.

The above correspondences between the edges and the vertices are one-to-one correspondences and these preserve the adjacency of vertices.

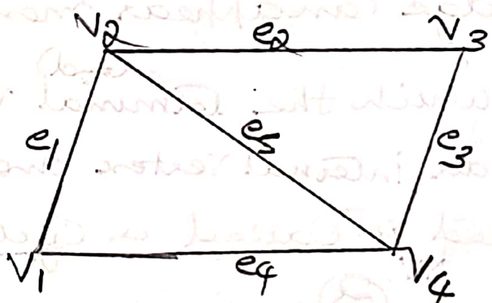
Hence the two graphs are isomorphic.

Definitions

Walk: Consider a finite, alternating sequence of vertices and edges of the form $v_i e_j v_{i+1} e_{j+1} v_{i+2} \dots e_k v_m$ in a graph G, which begins and ends with vertices such that each edges in the sequence is incident on the vertices preceding and following it in the sequence. Such a sequence is called a walk in G.

In a walk, a vertex or an edge (or both) can appear more than once.

The number of edges present in walk is called its length.



G

① The sequence $v_1 e_1 v_2 e_2 v_3 e_3 v_4$ is a walk of length 3. [no vertex and no edge is repeated]

② $v_1 e_1 v_2 e_5 v_4 e_3 v_3 e_2 v_2$

The sequence is a walk of length 4.
[vertex v_2 is repeated & no edge is repeated]

③ The sequence $v_2 e_2 v_3 e_3 v_4 e_5 v_2 e_5 v_4 e_4 v_4 e_1 v_2$ is a walk of length 6. [edge e_5 and vertices v_2 & v_4 are repeated]

A walk that begins and ends at the same vertex is a closed walk.

A walk that begins and ends at two different vertices is an open walk.

An open walk in which no edge appears more than once, is called a trail.

A closed walk in which no edge appears more than once is called a circuit.

A trail in which no vertex appears more than once is called a path.

(or)

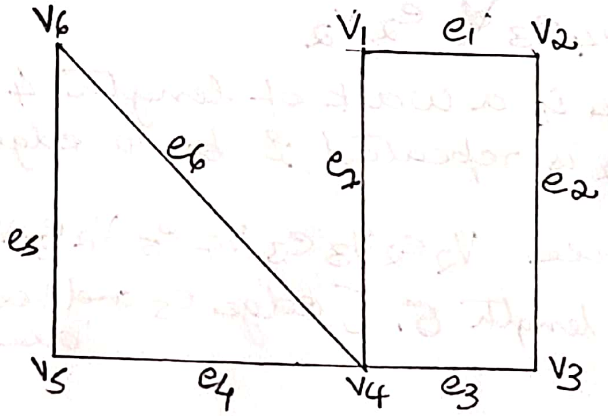
A path is an open walk in which neither a vertex nor an edge can appear more than once.

A circuit in which the terminal (end) vertex does not appear as an internal vertex and no internal vertex is repeated is called a cycle

(or)

A cycle is a closed walk in which neither a vertex nor an edge can appear more than once.

Ex:-



1) $v_4 e_7 v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5$ is a trail.

2) $v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_6 v_6$ is a path.

3) $v_6 e_5 v_5 e_4 \underline{v_4} e_3 v_3 e_2 v_2 e_1 v_1 e_7 \underline{v_4} e_6 v_6$ is a Circuit.

4) $v_6 e_6 v_4 e_4 v_5 e_5 v_6$ is a Cycle.

Points

1) A path with n vertices is of length $n-1$.
2) If a cycle has n vertices, it has n edges.
3) The degree of every vertex in a cycle is two.

5.5) Definition - Connected graphs

Two vertices in a graph G are said to be Connected if there is atleast one path from one vertex to the other.

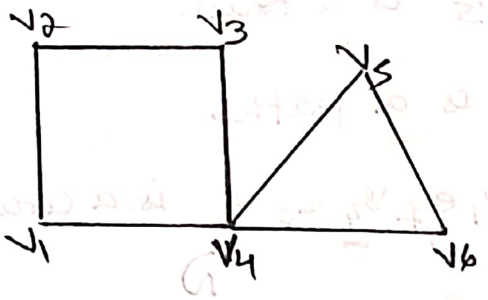
A Graph G is a Connected graph if every pair of distinct vertices in G are Connected.

or

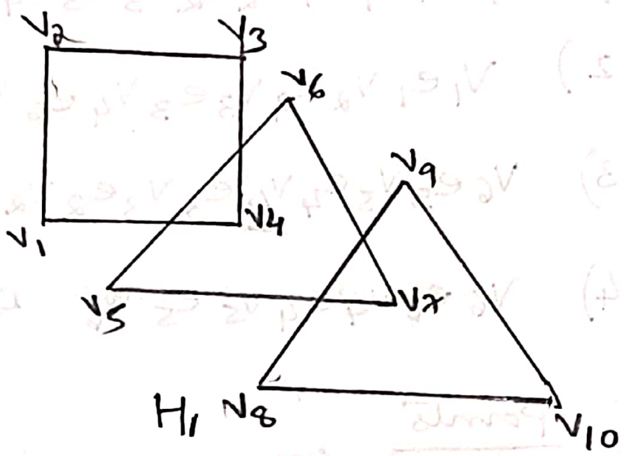
A Graph G is Connected if there is at least one path between every two distinct vertices of G.

A graph G is disconnected if G has at least one pair of distinct vertices between which there is no path.

Every graph G Consists of one or more Connected graphs. Each connected graph is a subgraph of G and is Called a Component of G.



G_1



H_1

A Connected graph has only one Component.

A disconnected graph has two or more Components.

The number of Components of a graph G is denoted by $k(G)$.

Ex: G_1 is a Connected graph.

H_1 is a disconnected graph, it has 3 Components.

i.e $k(H_1) = 3$

Theorem 1:- If a graph has exactly two vertices of odd degree, then there must be a path Connecting these vertices.

Proof:- Denote the two vertices of odd degree by v_1 and v_2. Suppose there is no path Connecting the vertices v_1 and v_2. Then the graph is disconnected

$\Rightarrow v_1 \in H_1$ and $v_2 \in H_2$, as H_1 & H_2 are two different Components.

Consequently, each of H_1 and H_2 Contains only one vertex of odd degree. This is not possible, because H_1 and H_2 graphs and in a graph the number of vertices of odd degree is always even.

Hence, there must be a path Connecting v_1 and v_2.

Theorem 4: A graph G is disconnected if and only if its vertex set V can be partitioned into 2 non-empty disjoint subsets V_1 and V_2 such that there exists no edge in G whose one end vertex is in V_1 and other is in V_2.

Proof :- Consider a disconnected graph G, and a vertex v in G. Let V_1 be the set of all vertices in G that are connected to v.

$\Rightarrow V_1$ does not include all vertices of G, as G is disconnected.

$\Rightarrow V_1$ is a proper subset of V

Let $V_2 = V - V_1 \Rightarrow V_1 \cap V_2 = \emptyset$ and $V_1 \cup V_2 = V$ and no vertex in V_1 is connected to any vertex in V_2.

Hence, V_1 and V_2 form a partition of V.

Conversely, suppose two subsets V_1 and V_2 of V form a partition of V. Consider 2 arbitrary vertices v and u in G. such that $v \in V_1$ and $u \in V_2$.

Then there exists no path between v and u.

Hence G is not connected.

Problems

1) If G is a simple graph with n vertices in which the degree of every vertex is atleast $(n-1)/2$, prove that G is connected.

Solution:- Consider any two vertices u & v in a simple graph G. Then they may be either adjacent or not adjacent. If they are adjacent, then G is connected. otherwise (not adjacent), each has atleast $(n-1)/2$ neighbours, because the degree of every vertex is atleast $(n-1)/2$. Therefore u and v taken together have atleast $n-1$ neighbours. Since G has a total of n vertices, the total number of neighbours which u and v together can have is only $n-2$.

Therefore, atleast one vertex, say x, is a neighbour of both u and v. Hence, there is an edge between u & x and x & v. Thus, there is a path between u & v. Hence G must be connected.

3. Let G be a disconnected graph of even order n with two components each of which is complete. Prove that G has a minimum of $n(n-2)/4$ edges.

Proof:- Let x be the number of vertices in one of the components of G. Then the other component has $n-x$ vertices. Since both components are complete graphs, the number of edges they have are $\frac{1}{2}x(x-1)$ and $\frac{1}{2}(n-x)(n-x-1)$ respectively.

Therefore, the total number of edges in G is m

ie $m = \frac{1}{2}x(x-1) + \frac{1}{2}(n-x)(n-x-1)$

$$= \frac{1}{2}[x^2 - x] + \frac{1}{2}[n^2 - nx - n - nx + x^2 + x]$$

$$= \frac{1}{2}[x^2 - \cancel{x} + n^2 - 2nx + x^2 - n + \cancel{x}]$$

$$= \frac{1}{2}[2x^2 - 2nx + n^2 - n]$$

$$m = x^2 - nx + \frac{1}{2}n(n-1) \quad \text{--- (1)}$$

$$\Rightarrow \frac{dm}{dx} = 2x - n, \quad \frac{d^2m}{dx^2} = 2 > 0$$

$\Rightarrow$ m is minimum, when $2x - n = 0 \Rightarrow x = n/2$

$\therefore$ Put $x = n/2$ in (1), Min $m = \left(\frac{n}{2}\right)^2 - n\left(\frac{n}{2}\right) + \frac{1}{2}n(n-1)$

$$= \frac{n^2}{4} - \frac{n^2}{2} + \frac{n^2}{2} - \frac{n}{2}$$

$$= \frac{1}{4}n(n-2)$$

Thus, the minimum number of edges in G is $\frac{n(n-2)}{4}$.

5.6 - Eulerian Graphs

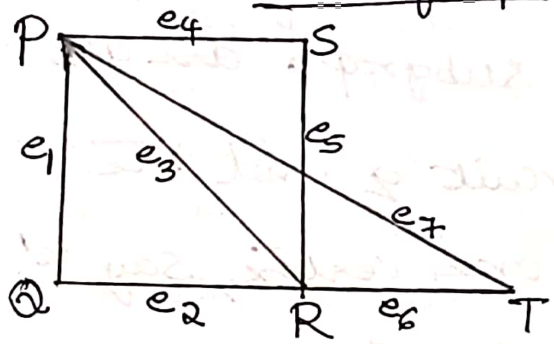
Definitions

If there is a circuit in a connected graph G, that contains all the edges of G, then that circuit is called an Euler circuit.

If there is a trail in G that contains all the edges of G, then that trail is called an Euler trail.

A connected graph that contains an Euler trail is called a semi-Euler graph.

A connected graph that contains an Euler circuit is called an Euler graph or Eulerian graph



G

$\Rightarrow$ The closed walk

$Pe_1Qe_2Re_3Pe_4Se_5Re_6Te_7P$

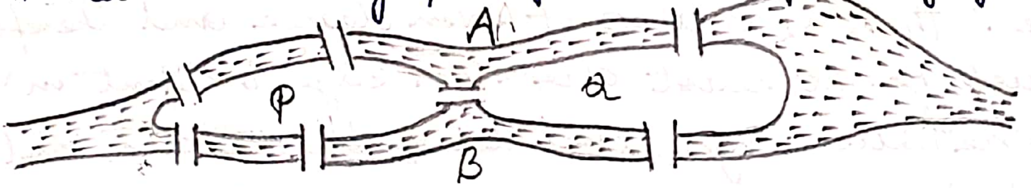
is an Euler circuit. Therefore the graph G is an Euler graph.

The Konigsberg Bridge Problem

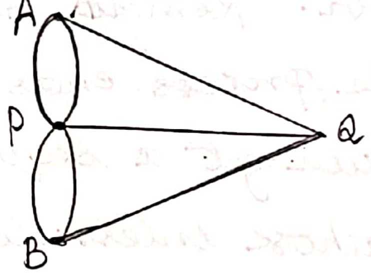
In the eighteenth century city named Konigsberg in East Prussia (Europe), there flowed a river named Piegel River which divided the city into four parts. Two of these parts were the banks of the river and two were islands. These parts were connected with each other through seven bridges.

The citizens of the city seemed to have posed the following problem. By starting at any of the four land areas, can we return to that area after crossing each of the seven bridges exactly once.

This problem, now known as the Konigsberg Bridge problem, remained unsolved for several years. In the year 1736, Euler analysed the problem with the help of a graph and gave the solution. This was indeed the starting point for the development of graph theory.



Let us see what the solution is as given by Euler. Denote the land areas of the city by A, B, P, Q, where A, B are the banks of the river and P, Q are the islands. Construct a graph by treating the four land areas as four vertices and the seven bridges connecting them as seven edges. The graph is as shown as below.



In this graph

$\deg(A) = \deg(B) = \deg(Q) = 3$

$\deg(P) = 5$

which are not even. Therefore, the graph does not have an Euler Circuit. This means that there does not exist a closed walk that contains all the edges exactly once. This amounts to saying that it is not possible to walk over each of the seven bridges exactly once and return to the starting point.

5.7 Hamiltonian graphs

Definition :-

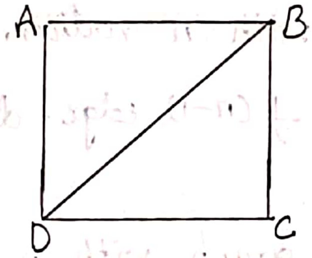
If there is a cycle in a connected graph G that contains all the vertices of G, then that cycle is called a Hamilton cycle in G.

A graph that contains a Hamilton cycle is called a Hamilton graph or Hamiltonian graph.

Ex :

ABCDA is a cycle in graph H.

∴ The graph H is a Hamilton graph.



graph-H

A path in a connected graph which includes every vertex of the graph is called a Hamilton path or Hamiltonian path in the graph.

Ex :

The path ABCD is a Hamiltonian path in the graph H.

Sufficient condition for a simple graph to be a Hamiltonian graph. [theorem-1]

Theorem 1

If in a simple connected graph with n vertices ($n \geq 3$) the sum of the degrees of every pair of non-adjacent vertices is greater than or equal to n, then the graph is Hamiltonian.

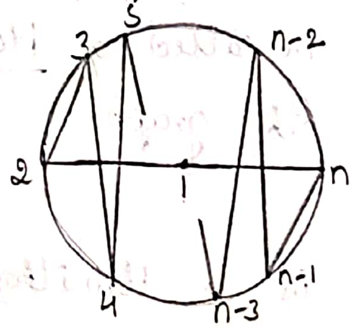
Theorem 2

If in a simple connected graph with n vertices ($n \geq 3$) the degree of every vertex is greater than or equal to $n/2$ then the graph is Hamiltonian.

Theorem 3.

In the Complete graph with n vertices, where n is an odd number ≥ 3, there are $\frac{1}{2}(n-1)$ edge-disjoint Hamiltonian cycles.

Proof: Let G be a complete graph with n vertices, where n is odd and $n \geq 3$. Denote the vertices of G by 1, 2, 3 --- n and represent them as points, as shown in figure.



we note that the polygonal pattern of edges from vertex 1 to vertex n as depicted in fig. is a cycle that includes all the vertices of G. This cycle is therefore a Hamilton cycle. This representation demonstrates that G has atleast one Hamilton cycle.

Now, rotate the polygonal pattern clockwise by $\alpha_1, \alpha_2 \cdots \alpha_K$ degrees, where $\alpha_1 = \frac{360}{n-1}$, $\alpha_2 = 2 \cdot \frac{360}{n-1}$, $\alpha_3 = 3 \cdot \frac{360}{n-1}$, ----

$$\cdots\ \alpha_K = k \cdot \frac{360}{n-1} = \frac{n-3}{2} \cdot \frac{360}{n-1}.$$

Each of these $k = (n-3)/2$ rotations gives a Hamilton cycle that has no edge in common with any of the preceding ones. Thus, there exists $k = (n-3)/2$ new Hamilton cycles, all edge-disjoint from the one shown in fig. and also edge-disjoint among themselves. Thus, in G, there are exactly

$$1 + k = 1 + \frac{n-3}{2} = \frac{1}{2}(n-1)$$

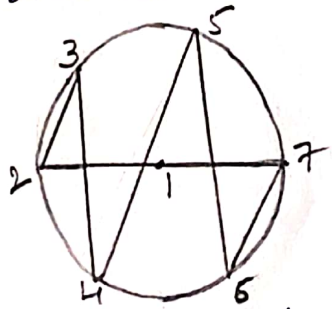
mutually edge-disjoint Hamilton cycles.

This completes the proof of the theorem.

Ex:-

1) How many edge-disjoint Hamilton cycles exist in the complete graph with seven vertices? Also, draw the graph to show these Hamilton cycles.

→ According to Theorem 3, the complete graph K_n has $\frac{1}{2}(n-1)$ edge-disjoint Hamilton cycles when $n \geq 3$ and n is odd. When $n = 7$, their number is $(7-1)/2 = 3$. As indicated in the proof of theorem 3, one of these Hamilton cycles appears as shown in fig.



The other two cycles are got by rotating the above shown cycle clockwise through angles $\alpha_1 = \frac{360^\circ}{7-1} = 60^\circ$ & $\alpha_2 = 2 \times \frac{360}{7-1} = 120^\circ$

(2) (a) Show that the number of Hamilton cycles in the complete bipartite graph $K_{n,n}$ (for $n \geq 2$) is $\frac{1}{2}(n-1)!\,n!$.

(b) How many different Hamilton paths are there in $K_{n,n}$?

→ (a) Let X and Y be the bipartites of $K_{n,n}$, with $X = \{x_1, x_2, \ldots x_n\}$ and $Y = \{y_1, y_2, \ldots\}$. Then each edge of $K_{n,n}$ is of the form $\{x_i, y_j\}$, $i, j = 1, 2 \ldots n$. Start with the vertex x_1. Since this vertex is on every Hamilton cycle of $K_{n,n}$, there are n choices for y_j, where $\{x_i, y_j\}$ belongs to the cycle. From y_j we have to return to X in n-1 ways for the second edge $\{y_j, x_i\}$ of the cycle where $i = 2, 3, \ldots n$. Continuing in this manner we obtain $(n-1)!\,n!$ number of cycles. Since directions are not involved, the total number of cycles, is half this number, namely $\frac{1}{2}(n-1)!\,n!$.

(b) Every Hamilton path in $K_{n,n}$ starts in either X or Y and terminates in the other. The number of such paths is $(n!) \times (n!) = (n!)^2$.